Blockchain & Cryptocurrency Exchanges

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Abstract

The use of blockchain technology has been pioneered by cryptocurrencies, namely Bitcoin. This technology has since proven to be a revolutionary concept, proving its usefulness in fields far removed from finances including logistics and manufacturing [15, 16]. Blockchain has since become a stalwart of other cryptocurrencies, developing into a critical component of cashless and decentralized currency systems. One of blockchain's greatest aspects is the immutability of the design. It forms a permanent record of all transactions, with each successive transaction being linked through references to the previous [17]. This security by design has been one of its major aspects that led to its meteoric rise to fame, as it makes the systems extremely difficult to manipulate. Still, blockchain technologies have been shown to be most vulnerable, and ideal for attack, in areas where different blockchains are "bridged" by various software, creating a diverse attack surface for attackers to circumvent the built-in security of blockchain. The use of bridging software has become the most targeted aspect of blockchains, resulting in $2 billion worth of assets stolen in 2022 alone [1]. This paper will explore the design and functionality of blockchain technology, how bridges function and how they can be exploited to steal cryptocurrency. The paper will also explore several solutions to address these security issues.

Introduction

Cryptocurrencies have become a significant financial asset, with the cryptocurrency market valued at $1.1 trillion as of August 2022. [2] Binance is currently the world's leading cryptocurrency exchange, averaging around $14 billion worth in exchanged assets in a 24-hour period and seeing over 15 million visitors in a 7-day period [3]. But it is by no means the only cryptocurrency exchange. There are currently 245 cryptocurrency exchanges that are actively monitored [3]. The value of cryptocurrency assets has made them a significant and formidable target for attackers.

As society continues to move toward a cashless system, technologies such as Apple Pay and Samsung Pay, which utilize Near Field Communications (NFC) technologies are increasingly implemented at points of sale. The increasing digitalization of currency and the infrastructure to support this has presented unique challenges, especially given that current currencies such as the dollar predated computer and internet technologies. Amongst the current generations, it is entirely possible to earn income, and spend and invest it without ever once handling a tangible physical representation [19].

Cryptocurrencies are entirely digital, lacking any physical form, and are built from the ground up in the internet era. As an entirely digital asset, they have been designed from conception to utilize computer networks globally via the internet. Consequently, the unique challenges of internet transactions have been factored in [23].

By design, cryptocurrencies are difficult to attack directly. Bitcoin was one of the first to leverage a technology known as blockchain. Since then, virtually all cryptocurrencies feature some form of blockchain technology to underpin their currency. Much has been improved since the original blockchain technology was developed, but even older designs remain remarkably resilient to attack [17].

Blockchain technology is a complex and interdisciplinary technology, incorporating aspects of cryptography, mathematics, and computer sciences to create the infrastructure. At its core, blockchain technology is an immutable ledger. Each transaction is considered a "block" which is connected to a "chain" of all previous transaction "blocks", hence the name, "blockchain." Within each block are references to previous blocks. These references make it virtually impossible to modify or delete blocks from the blockchain, because in doing so, the tampering would become immediately apparent unless the entire blockchain were modified. For this reason, the greater the number of blocks, the more secure a blockchain is [17].

The core functions and design of blockchain technology are beyond the scope of this paper. Suffice it to say, that direct blockchain attacks make up an extremely small number of the recorded attacks as of 2022, due to the sheer complexity and maturity of most blockchain systems today [4]. Instead, attackers have focused on the points of exchange where most of cryptocurrency assets are purchased, stored, and sold.

Cryptocurrencies such as Bitcoin were originally designed to utilize a Peer-to-Peer (P2P) type exchange, taking a completely decentralized approach to financing [23]. This approach is completely opposite of the current financial system, which utilizes central institutions to create, issue, maintain, and cull currency such as the US Federal Reserve or the European Central Bank. However, such a decentralized network also makes it difficult for people to interact with the system.

Cryptocurrency exchanges, then, emerged a solution to this issue. By setting their exchange as a node, or series of nodes, interacting with a blockchain network, they could act as a distributor of cryptocurrencies. A unique vulnerability is created; however, in exchanging national currencies for cryptocurrencies, or exchanging cryptocurrencies for other cryptocurrencies, because exchanges must use a software construct called a bridge.

Bridges have become the primary target for attackers because they represent the most vulnerable aspect of cryptocurrency exchange [1]. This paper will explore the design of bridges, how they can be exploited, and how some of the vulnerabilities might be addressed.

Background

Cryptocurrency bridges are created and implemented to bridge different blockchain systems. The bridges act much like a broker that facilitates an exchange of assets like stocks or bonds, setting up the necessary framework for an exchange to take place.

Much of the research for this paper centered on CAPEC-475, which is a vulnerability caused by signature spoofing with improper validation. This proved challenging to apply to the field of cryptocurrencies and blockchain. Although blockchain uses cryptographic signatures, the

encryption and validation methods used by blockchain make finding an effective attack very difficult. The number of scholarly sources investigating how a signature spoofing attack might be conducted on blockchain by exploiting an improper validation algorithm were virtually absent from the literature.

Overall, research continues to show the integrity and security of blockchain technology regarding cryptographic signatures and validations. However, it became clear that the exchanges where cryptocurrencies are exchanged often utilize proprietary software that has been exploited in significant heists over in the past three years. Attacks on these bridges were responsible for 69% of all cryptocurrencies stolen in 2022 [4], and current research identifies exchange bridges now as the primary target for attacks. This is predicted to continue for the foreseeable future [4].

Applying the scholarly research directly to the research focus proved challenging. Many aspects of cryptocurrency bridges utilize proprietary software and solutions. This makes public testing and evaluation of bridge designs difficult. Thus, finding examples of academic research investigating exchange bridge design were virtually non-existent. The overwhelming majority of results were media articles based on the official statements released from cryptocurrency exchanges post-attack. Still, some scholarly sources proved insightful for this paper.

Risk Analysis and Threat Model

The primary focus of this paper centers on the bridge frameworks. The design of the bridge varies between exchanges; however, the core design principles remain. Here the paper will explore the basic design of a cryptocurrency bridge, key vulnerabilities, and design challenges.

System functionality

At its core, a cryptocurrency exchange bridge is a facilitator between two different blockchains. Because two blockchains function as separate highways, a structure must be created to connect them. Bridges are the solution devised to resolve this issue [9].

There is significant diversity between bridge designs. Some of this diversity is due to the limitations imposed by the target cryptocurrency, that is, how the blockchain system is designed and determines with which methods it can be interfaced [9]. There are two main functional designs for cryptocurrency bridges.

The first is a unidirectional bridge. Unidirectional bridges allow exchange from one cryptocurrency into another, but the direction cannot be reversed, meaning once exchanged into the target cryptocurrency, there is no way to exchange back into the original cryptocurrency [9]. The second is a bidirectional bridge. Unlike unidirectional bridges, bidirectional bridges have no limitations on the number or ability to exchange cryptocurrencies between blockchains [9]. Naturally, of the two main designs, bidirectional bridges are the most functional.

If bridges themselves can be likened to physical structures bridging two highways, the method of transference they use can then be likened to how traffic moves across the bridge, except the type of vehicle must sometimes be changed before entering the next highway. By design, then,

bridges must create and utilize some form of token that becomes a compatible asset with the target blockchain [9]. Bridges have two main methods of how the this is performed. The first method is centralized, and the second method is decentralized.

In a centralized bridge, a central, standardized service creates and holds these token assets. For any exchange into that particular currency, all must go through this bridge. In a decentralized bridge, a local protocol within the exchange creates and holds the assets [9].

Naturally, there are proponents for both methods of asset transference. Centralized bridges are significant structures, designed as the central point for all exchanges into and out of a currency. This makes the structure a highly vetted and supported system because all exchanges must utilize it. However, it also makes it a singular target upon which an entire system depends. Any compromise to the structure could theoretically have catastrophic consequences.

With decentralized exchanges, exchange protocols are localized, and avoid the use of a singular, monolithic structure to conduct exchanges. The advantage with such designs rests in that they are compartmentalized. In the event there is a compromise to the structure, the damage only affects that one exchange. Additionally, the use of different protocols per exchange, makes targeting a singular attack vector by attackers more difficult. That is, each decentralized bridge often presents a unique attack surface area, which requires specialized attack designs. However, such decentralized exchanges are often fragmented in their approach to bridge design, utilizing proprietary, closed-source software. Without an external, 3rd-party certification or vetting system, or without the significant resources invested into central exchanges, any lapses or oversight in security can lead to significant or catastrophic failures. Currently, decentralized exchanges have incurred greater losses than centralized exchanges including the highly publicized Ronin Bridge Attack and Wormhole Bridge Attack just in 2022 [4, 20].

Regarding the specifics of the token creation and management protocol, it is both cryptocurrency specific, due to the limitations of a particular cryptocurrency, and bridge specific. There are several unique approaches to these design challenges; however, the predominate design methodologies can be placed in two primary categories: Smart Contract and Liquidity Pools [10].

With smart contract-type bridges, the original cryptocurrency asset is “wrapped” in another token with more universal acceptance and exchangeability. One common example of this is Wrapped Bitcoin (WBTC), which has been encapsulated inside of an ERC-20 token generated by the bridge. ERC-20 is the token protocol standard on the Ethereum network, which as become a de facto standard for a significant number of other cryptocurrency protocols as well [10]. Once encapsulated in an ERC-20 token, the Bitcoin asset (now a WBTC) can interact with most cryptocurrency blockchains to be exchanged [10].

Within smart contract-type bridges, there are three types: Burn and Mint, Lock and Mint, and Lock and Unlock. With Burn and Mint, the exchange bridge “burns” or destroys the asset on the original blockchain and mints a new asset from it on the target blockchain. With Lock and Mint, which is the most common design, the original assets are “locked” into a smart contract, which then mints new assets on the target blockchain. With Lock and Unlock, the original

cryptocurrency is locked on the original blockchain and the new exchange assets are unlocked from a liquidity pool on the target blockchain [12].

With liquidity pool-type bridges, the exchange itself manages a reservoir of various cryptocurrencies with which an exchange can be made. This approach is analogous to a digital bank, where one asset is exchanged at the current exchange rate for another asset. This reservoir of flexible assets to be used in an exchange is a pool of liquid or flexible assets, hence the term liquidity pool [10].

The basic design of a cryptocurrency exchange bridge is illustrated in *Figure 1* as a state diagram.

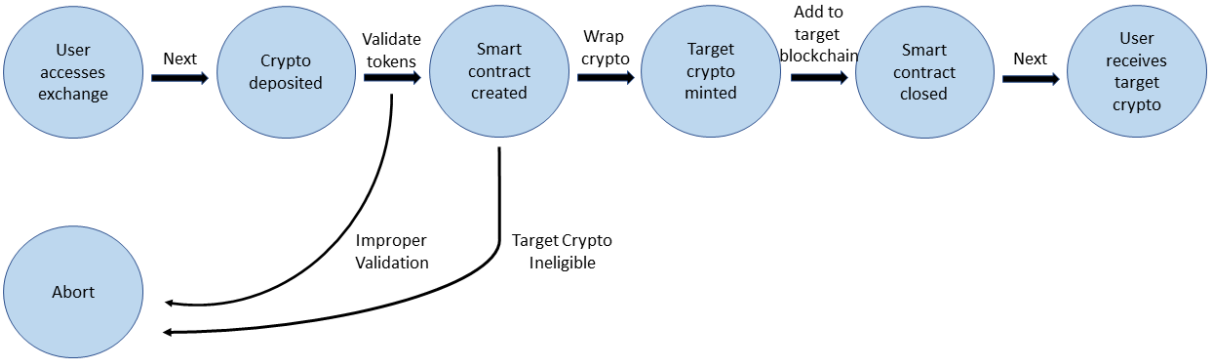

Figure 1-State transition diagram of a "Lock and Mint" cryptocurrency exchange bridge [10,11,12]

Threat Paths

In analyzing cryptocurrency bridges for threats, it is first helpful to assess how a user, and potentially an attacker, might interact with the exchange, and where exploits could be made. In *Figure 2*, a data flow diagram (DFD) is provided to illustrate the core architecture of a cryptocurrency exchange bridge.

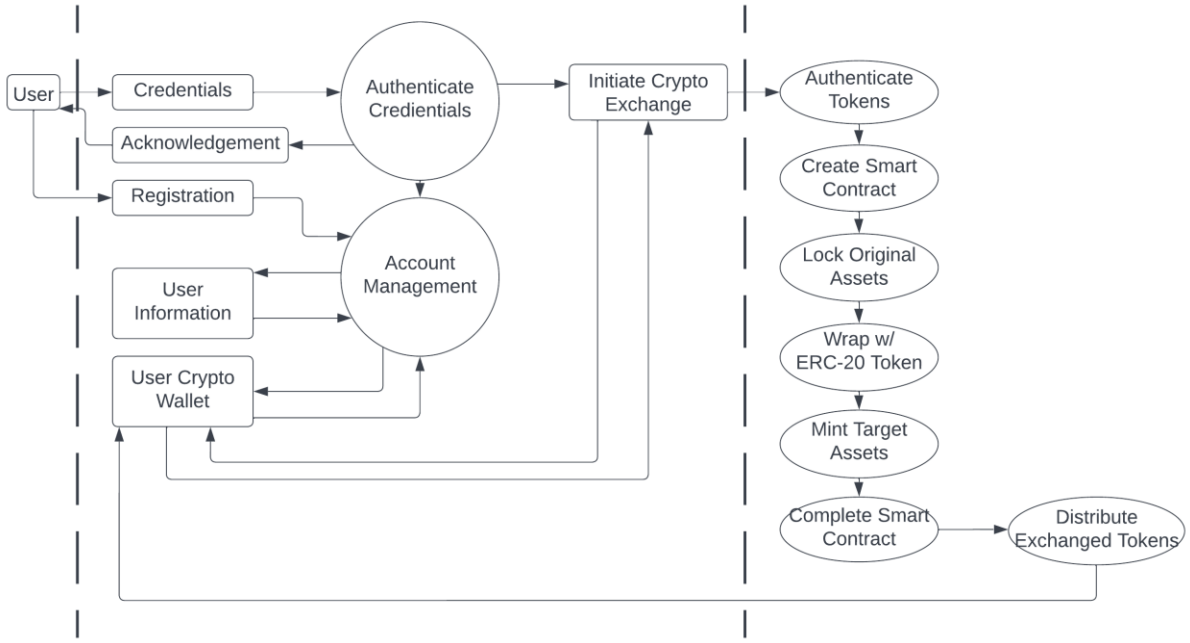

Figure 2-DFD illustrating the core architecture of a Lock and Mint secure contract cryptocurrency bridge [11-14]

The design of most cryptocurrency exchanges features user-friendly web-based design. Websites, with conventional accounts and logins are the most common way of interacting with a cryptocurrency exchange. As such, the standard user account and credentialing are an integral aspect of the initial interaction.

Linked to the user account is a “wallet” or “vault” with a unique address. It is here that a user’s cryptocurrencies are stored with the exchange, much like a safe deposit box or account would be with a bank. At any time, the user can withdraw these funds or deposit additional cryptocurrency funds. An extension of this is the ability to link a user’s bank account, which is considered in this diagram to fall under “User Information”. These national currency funds can be added to a crypto wallet to fund an exchange of national currency for a cryptocurrency later.

Once a user has created an account, or managed it as needed, it is then possible to commence a cryptocurrency exchange. In the DFD above, the architecture is based on the most common bridge design, the Lock and Mint design [11]. Once the user has funded the exchange wallet with a cryptocurrency, in this example, Bitcoin, the user may then choose to exchange the Bitcoin for another cryptocurrency. The first step is the authenticate the tokens to be exchanged, that they are validated correctly. Once the veracity of the Bitcoin tokens is confirmed, the exchange may begin.

The bridge software creates a smart contract, which locks the Bitcoin tokens, tying them up so they cannot be used elsewhere. The software then “wraps” them with another token, known as ERC-20, which allows the Bitcoin (now WBTC) to interact with a myriad of different

cryptocurrencies that support the ERC-20 token. The smart contract software then matches the value of the wrapped Bitcoin to the corresponding number of the desired cryptocurrency, in this case, Ether, and locks both into the contract. The contract is then complete, and the wrapped Bitcoin are then transferred with the Ether being sent to the user's crypto wallet.

Vulnerabilities and Attacks

Depending on the source, the number of main bridge architectures varies from three to five [11,12]. While the core underpinnings are similar, there is tremendous variance in design philosophies and methodologies. This creates both a wide variety of potential attack vectors, and the potential for unique, bridge specific attacks. For this reason, this paper focuses on Lock and Mint bridge architectures, which is the most used bridge architecture [12]. At its core, Lock and Mint bridge architectures hinge on smart contracts. These smart contracts are negotiated by bridge software, which are often written and maintained as proprietary software [22].

Scholarly research in this field remains surprisingly scant. It can be inferred, however, this dearth of research is due to the secrecy in which many cryptocurrency exchanges shroud their bridge software designs and code. Some of the scholarly research done this semester still provides interesting insight.

Pugh et al., explored the development of Post Quantum Cryptography (PQC) that is resistant to cracking by quantum and future supercomputers. In their research, it was shown that the rapid increase in system architectures and capabilities have already rendered earlier cryptographic algorithms obsolete and dangerous to use present-day. This illustrated the need for consistent reevaluation of cryptographic security measures [6]. In fields such as cryptocurrency exchange, the integrity and resilience of cryptographic algorithms is of paramount importance as attacks become more sophisticated. The Ronin Bridge attack exploited a weakness with cryptographic keys to interfere with the bridge's validation.

Liu et al. investigated proxy signature attacks, including forgery type attacks in their paper. They found that by hardening the discrete logarithmic problems, using a distribution scheme to share the incurred computational costs, they were able to produce a cryptographic signature framework that was significantly more resistant to the different types of signature attacks [7]. Building a cryptographic framework that is resistant to signature attacks is an essential aspect in exchanges.

Gattu et al., provided further unique insight on the use of side channel attacks, whereby placing an oscilloscope on the power supply to the computation center for encryption, the keys can be deduced algorithmically based on power signatures during the encryption process of a submitted plain text file [8]. Alternative methods of circumventing or defeating encryption processes should always be considered, especially when the entire functionality of a system is dependent on the protection encryption offers.

Moving from scholarly research, the design of a Lock and Mint bridge is vulnerable in several key points. The first is the token authentication. While the Ronin Bridge Attack utilized a unique aspect of the bridge design (the bridge utilized a majority consensus of 9 authenticator nodes), it leveraged a common aspect in bridges—authenticator nodes. Essentially eyewitnesses, these

nodes validate the authenticity of a transaction. If enough of these nodes are compromised, as with Ronin, or modified or bypassed, it is possible to steal cryptocurrency or crash the exchange.

Another aspect is the smart contract, where the currency to be exchanged is locked until the contract is complete. An attack here could unlock the cryptocurrencies tied up in the contract before the handoff is complete, rerouting them to another place. This year, Wormhole exchange experienced a vulnerability involving an uninitialized implementation of a contract owner [20].

Lastly, an attack post smart contract during the handoff of the newly exchanged currency could redirect the currency to a different location. Multichain exchange lost $1.4 million due to a logic vulnerability involving how funds were deposited [20].

Integrating Secure Testing

The first aspect of developing secure cryptocurrency exchange bridges is understanding the immense attack surface they now represent. With almost 70% of attacks now directed exclusively at exchange bridges, it has never been more important to build exchanges from the ground up with security in mind.

Future bridge designs should factor their attack surfaces into the architecture, focusing not just on effective design (does the bridge successfully exchange currencies between two blockchains?), but ensure that it is also secure. One of the ways this could be accomplished is leveraging new advancements in cryptography and secure software design. Da et al., investigated the use of cryptographic IP blocks using system-on-a-chip technology. By creating Physically Unclonable Functions (PUF), they were able to switch from private key management to Challenge Response Pairs (CRP) [21]. Testing the use of PUF cryptography with exchange bridges may prove useful for future designs and may have prevented the Ronin Bridge Attack from happening, since the attackers were able to exploit the private cryptographic keys to perform the attack.

Pugh et al, evaluated new and advanced encryption algorithms that will help encryption systems stay ahead of rapidly advancing technologies. To test the viability of new algorithms, Pugh et al. used the Bit Contribution Test (BCT), which tests to make sure it is not possible that two inputs produce the same a computed output. The implementation of advanced and improved encryption systems may help avoid issues where not just encryption strength is a concern, but also how the encryption framework is applied. This year alone, over $700 million in assets were lost to complications with private key management [20].

Finally, the collaborative development of an industry standard similar in design to ANSI (American National Standards Institute) would inform good design practices, helping exchanges to create secure bridges utilizing the best standards and practices, and certify bridge integrity.

Summary/Conclusion

The field of cryptocurrencies and their associated markets now represents over $1 trillion in financial assets. As the digitalization of financial systems and the rapid proliferation of online cryptocurrency exchanges continues, cryptocurrencies continue to grow in popularity and

applicability. As such, these centralized exchanges will clearly be a part of the cryptocurrency ecosystem for the foreseeable future. Within the exchanges, the bridges used to facilitate the exchanges should become a primary secure development focal point, with future bridge development focusing on the validator/authenticator nodes, management of private keys, and the development of collaborative industry standards and certifications.

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